scientific reports

Check for updates

OPEN

Knowledge, attitudes, and practices of parents regarding children's sun protection

Xuefei Lin[1]✉, Ting Li[2] & Xiao Chang[1]

This study provides the first comprehensive assessment of parental knowledge, attitudes, and practices regarding children's sun protection specifically in Beijing, addressing a regional evidence gap in China. This study aimed to investigate the knowledge, attitudes, and practices (KAP) of parents in Beijing regarding sun protection for their children. A cross-sectional survey was conducted at Xuanwu Hospital, Capital Medical University, between May 25 and June 30, 2025. Data were collected using a structured questionnaire, which included items on demographic characteristics and KAP-related variables. A total of 477 valid responses were obtained, with 357 respondents (74.84%) being mothers. Physical sun protection methods were preferred by 273 participants (57.23%). Among the children, 249 (52.20%) were female, 277 (58.07%) had an average daily sun exposure of 1–2 h, and 97 (20.34%) had experienced sunburn. The scores for knowledge, attitude, and practice were 6.256 ± 1.34 (range: 0–11), 30.95 ± 4.11 (range: 8–40), and 31.05 ± 7.67 (range: 10–50), respectively. Path analysis revealed that knowledge directly influenced practice (β = 0.52, *P* = 0.027), whereas attitude exhibited a stronger influence on practice (β = 0.74, *P* < 0.001). Parents in Beijing exhibited limited knowledge, generally positive attitudes, but suboptimal practices regarding sun protection for their children. The significant associations between knowledge, attitude, and practice suggested that targeted educational interventions aimed at increasing parental awareness and shifting attitudes could enhance sun protection behaviors in both clinical and community settings.

Keywords Sunlight, Sunscreen, Parental sun safety, Pediatric photoprotection, Behavioral determinants, Health behavior, Cross-Sectional study

Abbreviations

KAP	Knowledge, Attitudes, and Practices
UV	Ultraviolet
SPF	Sun Protection Factor
PA	Protection Grade of UVA
OR	Odds Ratio
CI	Confidence Interval
SD	Standard Deviation
ANOVA	Analysis of Variance
RMSEA	Root Mean Square Error of Approximation
IFI	Incremental Fit Index
TLI	Tucker-Lewis Index
CFI	Comparative Fit Index
UVA	Ultraviolet A
UVB	Ultraviolet B

Exposure to ultraviolet (UV) radiation during childhood is a significant concern in preventive dermatology, as the cumulative UV exposure during this period substantially contributes to lifetime risk of skin photoaging and carcinogenesis. According to global epidemiological data, approximately 132,000 cases of melanoma and more than 2.0 million cases of other types of skin cancer are reported worldwide each year, and skin cancer affects one

[1]Department of Dermatology and Venereal Disease, Xuanwu Hospital Capital Medical University, Beijing 100053, China. [2]Department of Dermatology and Venereal Disease, Xiong'an Xuanwu Hospital, Xiong'an New Area 070001, Hebei, China. ✉email: lxf881215@126.com

of three patients diagnosed with cancer[1]. Multiple epidemiological investigations have confirmed that excessive and unprotected UV exposure during childhood is associated with increased risk of all skin cancer types, with childhood sunburn being a particularly strong risk factor for melanoma development[2]. While the incidence of skin cancer remains lower in China than in countries like Australia and New Zealand, evidence suggests that skin cancer incidence is increasing in China[3]. In Chinese cities such as Guangzhou, year-round UVR intensity is approximately 84.3% above the "moderate" level, with 65.1% at "high" and "very high" levels, creating substantial exposure risks for children that warrant proper protection measures[4].

Parents and caregivers play a pivotal role in safeguarding children from harmful UV exposure through both direct intervention and behavioral modeling. A German nationwide study found that caregivers who perceived themselves as sun safety role models were significantly more likely to engage in protective behaviors for both themselves and their children, including a 5.08-fold increased likelihood of using sunscreen on their face and a 7.22-fold increase for their body[2]. Despite this understanding, studies have revealed significant deficiencies in parental knowledge and practices. Several studies have reported that misconceptions about sunscreen necessity and improper application techniques are widespread among Chinese parents, particularly regarding reapplication frequency and coverage areas[4]. Additionally, a review of 50 websites on children's sun safety found that most failed to emphasize the long-term benefit of sunscreen in significantly reducing risk for skin cancer and did not address key misconceptions about sunscreen safety[5].

The Knowledge, Attitude, and Practices (KAP) survey approach provides a robust framework for diagnostic research in health behavior, based on the premise that knowledge influences attitudes, which in turn shape behaviors[6,7]. In the context of children's sun protection, this methodology enables comprehensive assessment of parents' awareness of UV-related risks, their attitudes toward protective measures, and the actual preventive behaviors they implement in daily life. Given that Beijing has experienced increasing urbanization with corresponding changes in lifestyle and outdoor activity patterns among children, understanding parental KAP becomes particularly crucial for developing contextually appropriate interventions.

While research on children's sun protection has been conducted in various regions including Australia, where studies show that 69–85% of parents report regular sunscreen use for children[8], the United States, where parents consider sunscreen the most common method of sun protection[5], and Southern China[4], specific data pertaining to parental knowledge, attitudes, and practices in Beijing remains limited. Beijing's distinct climate, characterized by hot summers with high UV indices averaging above 8 during summer months, along with its urban environment and evolving lifestyle patterns, presents unique contextual factors that may influence parental approaches to sun protection. Empirical studies reinforce the pivotal role of parental behavior in shaping children's sun protection habits. In a population-based study in Guangzhou, parents who used sun protection measures themselves were significantly more likely to ensure their children were protected, with hats and umbrellas being the most commonly employed methods, while sunscreen use remained inconsistent[4]. Similarly, a large cross-sectional study in Germany revealed that inadequate sun protection in children was strongly linked to parental behaviors and household characteristics; children whose parents did not consistently use one protective method (e.g., hats or sunscreen) were significantly less likely to be protected by others, demonstrating a clustering effect in sun-safe behaviors[9]. Chinese children typically have Fitzpatrick skin types II, III, and IV, which differs from predominantly white populations studied in Western research and may influence protection practices[10]. Furthermore, existing studies from other Chinese regions suggest substantial knowledge gaps, with television advertisements (52.0%), newspapers (37.6%), and magazines (37.0%) being the primary sources of sun protection information rather than healthcare professionals (10.0%)[4]. Despite existing studies from Guangzhou and other Asian regions, no research has specifically examined parental sun protection behaviors within Beijing's unique climatic, urban, and sociocultural context, leaving an important geographic and behavioral gap in the literature. This study addresses this research gap by comprehensively assessing the knowledge, attitudes, and practices of parents in Beijing regarding children's sun protection. We hypothesized that parents in Beijing would demonstrate limited knowledge, generally positive attitudes, and suboptimal sun protection practices, and we aimed to quantitatively examine the relationships among these domains.

Methods

Study design and participants

This cross-sectional study was conducted at Xuanwu Hospital, Capital Medical University, between May 25 and June 30, 2025. The study population comprised parents of children, and all procedures were approved by the Ethics Committee of Xuanwu Hospital, Capital Medical University (MR-11-25-047140). Written informed consent was obtained from all participants prior to data collection.

Eligible participants met the following inclusion criteria: (1) being a parent or legal guardian of a child aged between 6 months and 16 years; (2) currently residing in Beijing; and (3) having the capacity to complete the survey independently. Parents of children with UV-related skin diseases, including photodermatitis, porphyria, and vitiligo, were excluded.

Questionnaire introduction

The questionnaire was developed with reference to relevant guidelines and existing literature[4,11–14], and its initial structure was evaluated through a pilot study involving 40 participants. These 40 participants were part of a pilot test only and were excluded from the final study sample. The instrument demonstrated high internal consistency, with an overall Cronbach's α coefficient of 0.894.

The finalized questionnaire, administered in Chinese, consisted of four main sections: (1) demographic information; (2) knowledge; (3) attitudes; and (4) practices. The demographic section captured key characteristics of the participants and their children, including the parent's age, relationship to the child, place of residence,

education level, and monthly household income per capita, as well as the child's age, gender, preferred sun protection methods, average daily sun exposure on regular days and holidays, and any prior history of sunburn.

The knowledge section comprised 11 items covering two domains. Responses were dichotomously scored, with correct answers receiving 1 point and incorrect answers receiving 0 points, yielding a total possible score ranging from 0 to 11. The attitude section included 8 items assessed using a five-point Likert scale, ranging from "strongly agree" (5 points) to "strongly disagree" (1 point), with a cumulative score ranging from 8 to 40. The practice section contained 9 items; among these, items 1 and 4 through 7 were rated on a five-point scale reflecting the frequency or quality of sun protection behaviors, from "very positive" (5 points) to "very negative" (1 point), resulting in a total possible score of 10 to 50. Attaining scores above 70% of the maximum in each section indicated adequate knowledge, positive attitude, and proactive practice[15]. The remaining items in the practice section were analyzed descriptively due to their categorical or multiple-choice formats.

Questionnaire distribution and quality control

Participants were recruited from the dermatology and cosmetic dermatology outpatient clinics and included both patients and their accompanying acquaintances. During routine consultations, individuals were asked whether they were parents of children within the eligible age range. Those who met the inclusion criteria and expressed willingness to participate were invited to a private consultation room following their medical appointment. There, they received a QR code granting access to the electronic questionnaire.

A dermatology attending physician provided standardized instructions on how to complete the questionnaire and was available to address any questions or clarify potential misunderstandings. To ensure data quality, responses submitted in under 90 s were deemed invalid. This threshold was based on findings from the pilot test, which indicated that questionnaires completed this quickly were unlikely to reflect careful reading or thoughtful answering; therefore, such entries were excluded from the analysis.

Sample size calculation

The required sample size was calculated using the formula:

$$\mathrm{n} = \left(\frac{Z_{1-\alpha/2}}{\delta}\right)^2 \times p \times (1-p)$$

n represents the required sample size per group.

α represents the Type I error, set at 0.05.

$Z_{1-\alpha/2}$ is obtained from statistical tables, with a value of 1.96 in this study.

δ represents the allowable margin of error, set at 0.05.

p is set at 0.5 (as this maximizes the required sample size).

n= $\left(\frac{1.96}{0.05}\right)^2 \times 0.5 \times (1-0.5) \approx 384$.

To account for potential non-response and ensure sufficient power, an expected effective response rate of 80% was applied. Accordingly, the final target sample size was adjusted to 480 distributed questionnaires. This sample size estimation was based on proportions for descriptive survey purposes and was not intended as a sample size calculation for path analysis. Because the path model was fully saturated (df = 0), formal SEM-specific sample size requirements did not apply, and the final sample of 477 participants was fully adequate for estimating the path coefficients.

Statistical methods

Data analysis was conducted using STATA 17.0. Continuous variables were expressed as means and standard deviations (SD), and categorical variables were presented as frequencies and percentages [n (%)]. Normality of continuous data was assessed using the Kolmogorov–Smirnov test. For comparisons between two groups, the independent samples t-test was used for normally distributed variables, while the Wilcoxon Mann–Whitney test was applied for non-normally distributed data. For comparisons among three or more groups, analysis of variance (ANOVA) was employed for data with normal distribution and homogeneity of variances, whereas the Kruskal–Wallis test was used for non-normally distributed data. Spearman correlation analysis was conducted to explore associations among knowledge, attitude, and practice scores. Multivariate logistic regression analysis was performed to identify factors independently associated with sun protection behaviors, using the top 70% of practice scores as the cut-off value. Path analysis was used to examine the direct and indirect pathways linking parental knowledge and attitudes to sun protection practices. The analysis was theoretically grounded in the KAP framework, which specifies directional relationships among these domains. Because the path model was fully saturated (degrees of freedom = 0), conventional model fit indices such as RMSEA, CFI, and TLI were not applicable or informative. In a saturated model, the number of estimated parameters equals the number of observed variances and covariances, resulting in zero degrees of freedom; consequently, chi-square–based statistics and derived indices default to RMSEA = 0, CFI = 1, and TLI = 1, which hold no evaluative value. Given that all parameters were identifiable under this saturated structure, no SEM-specific sample size assumptions were required, and the final sample of 477 participants was more than adequate for estimating the path coefficients. A two-sided P-value of less than 0.05 was considered statistically significant.

Results

Demographic information on participants

Initially, a total of 505 participants were recruited. Two individuals declined to participate, leaving 503 responses. The following responses were excluded: A total of 17 responses with a completion time of less than 90 s were excluded; Ten responses where the participant indicated they were not the child's parent. After exclusions, 477 valid responses remained, with an effective rate of 94.83%. Among the participants, 357 (74.84%) were mothers, with mean age of 38.34 ± 4.98 years, 275 (57.65%) had associate/bachelor's degree, and 273 (57.23%) preferred physical protection for sun protection. Among the children, 249 (52.2%) were girls, with mean age of 8.15 ± 3.98 years, 277 (58.07%) had an average daily sun exposure of 1–2 h, 97 (20.34%) experienced sunburns. The knowledge, attitude, and practice scores were 6.256 ± 1.34 (possible range: 0–11), 30.95 ± 4.11 (possible range: 8–40), and 31.05 ± 7.67 (possible range: 10–50), respectively. Analyses of demographics show that mothers had higher attitude scores and fathers had higher practice scores. Parents of children with longer sun exposure had higher attitude scores. In addition, parents of girls and parents who used a combination of measures to protect their children from sunburn had higher attitude and practice scores ($P < 0.05$) (Table 1).

		Knowledge score		Attitude score		Practice score	
	N (%)	Mean ± SD	*P*	Mean ± SD	*P*	Mean ± SD	*P*
Total score		6.256 ± 1.34		30.95 ± 4.11		31.05 ± 7.67	
Age (years old)	38.34 ± 4.98						
Relationship with the child			0.183		**0.030**		**< 0.001**
Father	120 (25.16)	6.15 ± 1.53		30.21 ± 4.34		33.24 ± 7.91	
Mother	357 (74.84)	6.29 ± 1.27		31.19 ± 4.00		30.30 ± 7.45	
Residence			**0.036**		0.162		**0.027**
Rural	7 (1.47)	7 ± 1.41		29.42 ± 5.99		38.14 ± 11.6	
Urban	459 (96.23)	6.26 ± 1.33		31.04 ± 4.04		31.02 ± 7.56	
Suburban	11 (2.31)	5.45 ± 1.43		27.90 ± 4.59		27.45 ± 6.90	
Education			0.730		0.500		0.317
High school/vocational school or below	34 (7.13)	6.35 ± 1.15		31.05 ± 4.50		32.52 ± 10.7	
Associate/Bachelor's degree	275 (57.65)	6.24 ± 1.35		31.18 ± 4.08		31.22 ± 7.54	
Master's degree or above	168 (35.22)	6.25 ± 1.35		30.55 ± 4.05		30.45 ± 7.10	
Monthly income per capita			0.593		0.703		0.685
< 5000	20 (4.19)	6.25 ± 1.11		31.55 ± 3.80		30.95 ± 11.3	
5000–10,000	100 (20.96)	6.17 ± 1.21		30.66 ± 3.85		30.66 ± 7.75	
10,000–20,000	165 (34.59)	6.2 ± 1.41		30.82 ± 4.08		30.98 ± 7.65	
> 20,000	192 (40.25)	6.34 ± 1.36		31.15 ± 4.29		31.30 ± 7.21	
Child's age [0 ~ 16]	8.15 ± 3.98						
Child's gender			0.527		**0.009**		**< 0.001**
Boy	228 (47.8)	6.24 ± 1.40		30.37 ± 4.19		29.60 ± 7.80	
Girl	249 (52.2)	6.26 ± 1.28		31.47 ± 3.96		32.36 ± 7.31	
Preferred sun protection method			0.147		**< 0.001**		**< 0.001**
Physical protection	273 (57.23)	6.31 ± 1.32		30.87 ± 3.79		30.88 ± 7.48	
Sunscreen spray	42 (8.81)	6.57 ± 1.41		31.5 ± 4.06		32.47 ± 7.02	
Combination of physical protection and sunscreen	123 (25.79)	6.09 ± 1.27		32.21 ± 3.83		32.97 ± 6.43	
No sun protection measures	39 (8.18)	6 ± 1.50		26.94 ± 4.62		24.53 ± 9.57	
Child's average daily sun exposure on regular days			0.791		**0.042**		0.241
< 1 h	75 (15.72)	6.22 ± 1.36		29.94 ± 4.46		29.89 ± 8.27	
1–2 h	277 (58.07)	6.22 ± 1.27		31.07 ± 3.97		31.14 ± 7.51	
> 2	125 (26.21)	6.35 ± 1.46		31.28 ± 4.12		31.51 ± 7.63	
Child's average daily sun exposure during holidays			0.136		0.091		0.398
< 1 h	66 (13.84)	5.92 ± 1.54		30.04 ± 3.79		29.90 ± 8.89	
1–2 h	211 (44.23)	6.23 ± 1.27		31.02 ± 3.98		31.56 ± 8.01	
2–4 h	160 (33.54)	6.44 ± 1.32		31.08 ± 4.32		30.8 ± 7.03	
> 4 h	40 (8.39)	6.15 ± 1.31		31.55 ± 4.32		31.15 ± 5.86	
Child experienced sunburn			0.961		0.058		0.454
Yes	97 (20.34)	6.22 ± 1.31		31.63 ± 3.73		31.71 ± 7.81	
No	380 (79.66)	6.26 ± 1.34		30.77 ± 4.18		30.87 ± 7.63	

Table 1. Demographic characteristics and KAP scores.

Distribution of responses to knowledge, attitude, and practice

The distribution of responses for the knowledge dimension showed that 15.72% of participants incorrectly believed that the skin will not get sunburned or tanned on cloudy days (K6), and 32.29% incorrectly believed that the higher the SPF and PA index, the better the sunscreen for children (K9). Notably, only 22.22% of participants agreed with the false statement that "sunscreen does not need to be reapplied regularly; a single application before outdoor activities is sufficient" (K10), while 77.78% correctly rejected this claim, indicating a good general awareness of the need for reapplication (Table S1).

Responses to the attitude dimension showed that 25.16% strongly agreed and 45.49% agreed that they are very concerned about various skin damage caused by sun exposure in children (A2), 5.66% strongly agreed and 18.03% agreed that frequent sun protection can lead to calcium deficiency in children (A4), and 4.61% strongly agreed and 15.3% agreed that as long as a child does not get sunburned, sun protection is unnecessary (A3) (Table S2).

Responses to the practice dimension showed that only 19.29% always took sun protection measures for their child (P1). Meanwhile, 54.51% take sun protection measures to their child 30 min before sun exposure, while the rest take measures after sun exposure (P3). Also, only 12.16% always reapply sunscreen to their children during prolonged sun exposure (P7). In addition, only 16.35% were concerned about sun protection on the torso (P8) (Table S3).

Spearman correlation analysis

Spearman correlation analysis indicated significant positive correlations between knowledge and attitude ($r = 0.098$, $P = 0.032$), as well as practice ($r = 0.116$, $P = 0.011$). Meanwhile, there was also correlation between attitude and practice ($r = 0.411$, $P < 0.001$) (Table 2).

Multivariate analysis

Multivariate logistic regression showed that attitude score (OR = 1.19, 95% CI: [1.12,1.26], $P < 0.001$), as mother (OR = 0.39, 95% CI: [0.24,0.64], $P < 0.001$), master's degree or above (OR = 0.35, 95% CI: [0.15,0.84], $P = 0.019$), as parents of girls (OR = 1.53, 95% CI: [1.00,2.35], $P = 0.049$) were independently associated with practice (Fig. 1).

Path analysis

As the model was fully saturated (df = 0), all relationships among variables were exactly reproduced, meaning that traditional model fit indices such as RMSEA, CFI, and TLI were not meaningful for evaluation. The results of path analysis showed that knowledge directly affected practice ($\beta = 0.52$, $P = 0.027$), attitude directly affected practice ($\beta = 0.74$, $P < 0.001$). Although the indirect effect of knowledge on practice through attitude did not reach statistical significance ($\beta = 0.18$, $P = 0.08$), knowledge remained significantly associated with practice in the path analysis model (Table 3).

Discussion

This study revealed a clear and consistent pattern in parental sun protection behaviors in Beijing, characterized by limited knowledge, generally positive attitudes, and inadequate protective practices. Parents in Beijing demonstrated insufficient knowledge, relatively positive attitudes, and suboptimal practices concerning sun protection for children.

Studies in this field have suggested that sun protection behaviors in children are shaped by a complex interplay of parental knowledge, attitudes, and actual practices, which are themselves influenced by broader socio-cultural, educational, and systemic factors. This study builds upon that framework by exploring how these dimensions manifest among parents in Beijing, revealing a landscape where awareness exists in fragments, attitudes are largely positive yet inconsistently acted upon, and practices remain selective and conditional. These findings underscore a wider pattern observed in other urbanized but transitional healthcare settings, where health knowledge does not seamlessly translate into behavior[16].

Parental knowledge in this sample demonstrated a generally moderate level of understanding regarding sun protection principles, although certain misconceptions remained. While many participants were aware that children are more vulnerable to ultraviolet (UV) radiation and acknowledged the importance of sun protection, their comprehension of specific practices varied across items. Encouragingly, 77.78% of respondents correctly rejected the false claim that sunscreen does not need to be reapplied regularly (K10), indicating a relatively strong awareness of the necessity of reapplication. However, misconceptions persisted in other areas: for example, 15.72% incorrectly believed that sunburn does not occur on cloudy days (K6), and 32.29% believed that higher SPF and PA values automatically mean better protection for children (K9). These results suggest that although general awareness exists, some critical details remain misunderstood, particularly regarding product selection and weather-related UV risks. It is also important to note that the correlation between knowledge and attitude was statistically significant but very small in magnitude ($r = 0.098$), indicating limited clinical or

	Knowledge	Attitude	Practice
Knowledge	1		
Attitude	0.098 ($P = 0.032$)	1	
Practice	0.116 ($P = 0.011$)	0.411 ($P < 0.001$)	1

Table 2. Correlation analysis.

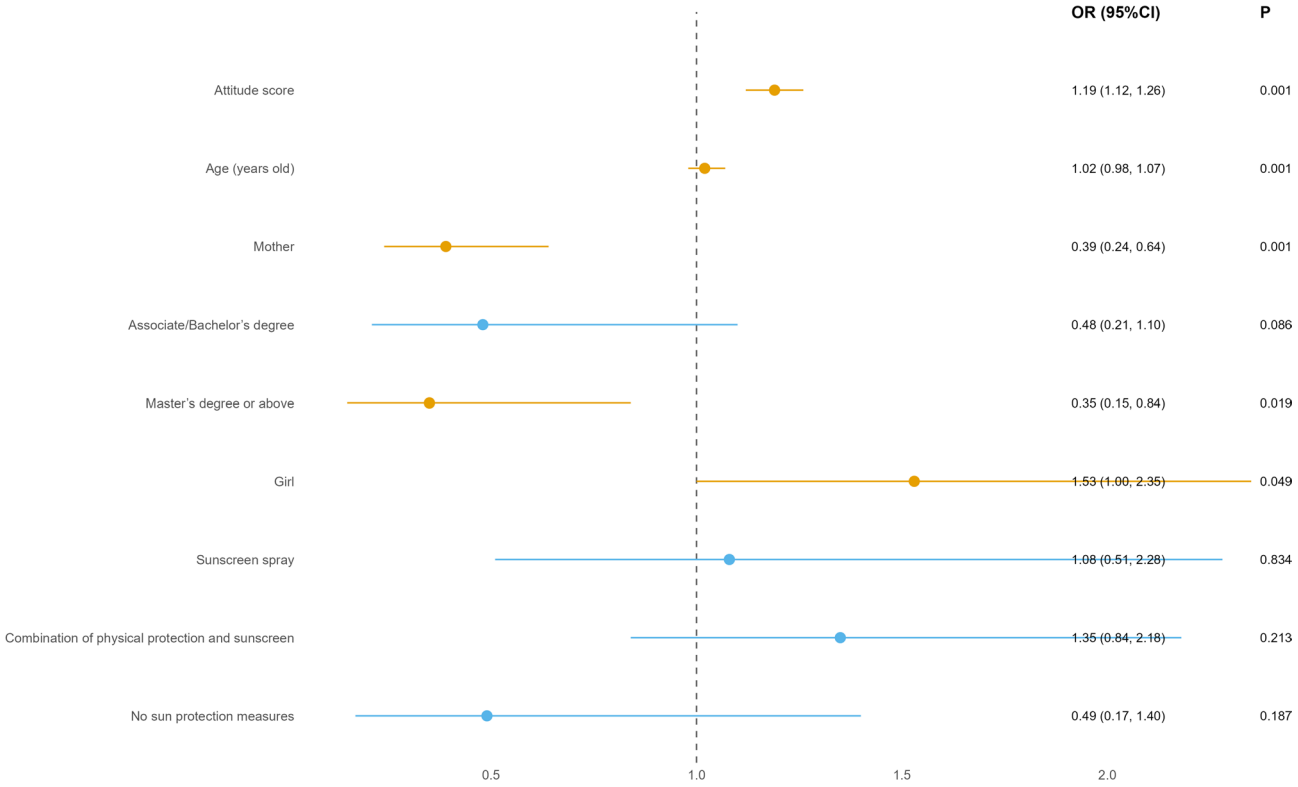


Fig. 1. Multivariate analysis for practice.

Model paths		Total effects		Direct Effect		Indirect effect	
		β (95% CI)	*P*	β (95% CI)	*P*	β (95% CI)	*P*
Asum ←							
	Ksum	0.24 (−0.02,0.52)	0.075	0.24 (−0.02,0.52)	0.075		
Psum ←							
	Asum	0.74 (0.59,0.90)	< 0.001	0.74 (0.59,0.90)	< 0.001		
	Ksum	0.71 (0.20,1.22)	0.006	0.52 (0.06,0.99)	0.027	0.18 (−0.02,0.39)	0.08

Table 3. Path analysis model (saturated model).

practical relevance despite its statistical significance. This highlights the need for more nuanced educational efforts targeting these gaps. Furthermore, marketing strategies that focus heavily on SPF ratings but fail to clarify the need for reapplication likely reinforce the misconception that a single application ensures all-day protection. Cultural beliefs and gendered expectations further compound these gaps. For example, the data show that parents of girls had significantly higher attitude and practice scores than those of boys, likely reflecting aesthetic concerns over tanning and pigmentation that are more socially emphasized for female children in Chinese contexts. Such motivations, while prompting greater vigilance, also risk narrowing the rationale for sun protection to appearance alone rather than long-term skin health. To address these issues, interventions must go beyond awareness campaigns and focus on debunking specific misconceptions. For example, pediatricians and community health workers should emphasize that UV radiation penetrates clouds and that reapplication every two hours is critical, even when using high-SPF products. Interactive education—such as brief videos, visual aids, or Q&A sessions in clinics—may improve comprehension and retention. Moreover, educational programs should consciously target both mothers and fathers, as fathers in this study demonstrated higher practice scores despite lower attitude scores, indicating that practical involvement does not always correlate with self-reported concern, and both parents should be actively engaged[4,9]. Misconceptions were particularly common in areas such as sunscreen application frequency, the meaning and role of SPF ratings, and the false belief that calcium deficiency may result from frequent sun avoidance. This aligns with findings from other studies in similar socio-economic contexts, which report that even among well-educated populations, public health messages often fail to achieve conceptual clarity or behaviorally relevant impact[17]. Notably, knowledge scores in our study did not vary significantly across income or educational groups, suggesting that informational outreach may be broadly distributed but shallow in depth. These results suggest a saturation of basic messaging but a lack of focused,

behavior-oriented education, particularly on topics requiring nuanced understanding, such as reapplication strategies and protection on overcast days.

Importantly, knowledge alone was not sufficient to drive consistent behaviors, and its influence on practice appeared to be context-dependent rather than operating as a stable independent determinant. In the multivariate logistic regression model, knowledge was not an independent predictor of sun protection practice after adjusting for parental role, education level, child's gender, and other covariates. This suggests that the apparent effect of knowledge may be attenuated or confounded when broader sociodemographic and behavioral factors are simultaneously considered. In contrast, the path analysis focused exclusively on the theoretical relationships among knowledge, attitudes, and practices without external covariates, within which knowledge demonstrated a significant direct association with practice[18,19]. These findings underscore the need for cautious interpretation of the direct effect of knowledge and suggest that improving attitudes and contextual support may be more critical for translating knowledge into consistent protective behaviors. This pattern is consistent with a cognitive–affective framework, in which factual understanding may shape health beliefs that subsequently guide action. However, given the very weak correlation between knowledge and attitude (r = 0.098), the indirect effect observed in the path analysis should be interpreted cautiously and may not represent a robust or clinically meaningful relationship[20]. This gap between knowing and doing signals the need for communication strategies that move beyond awareness campaigns, encouraging reflective engagement and personal risk assessment among caregivers.

Attitudinal responses in our study were generally positive, which reflects a promising foundation for behavior change. However, certain beliefs remain fragile and easily influenced by misinformation. A considerable proportion of parents expressed uncertainty or agreement with incorrect statements regarding calcium metabolism, and others held the belief that protection is only necessary when sunburn is visible. This reactive orientation to sun protection has been reported in previous Chinese studies, where parents often regard visible damage as the threshold for intervention[21]. Attitude emerged as the strongest determinant of behavior in both regression and path analyses, suggesting that strengthening this domain could have the greatest downstream impact on practice. Mothers in particular showed more favorable attitudes than fathers, echoing patterns from previous research on gendered caregiving roles in preventive child health[22]. The gender of the child also influenced parental behavior, with parents of girls more likely to engage in protective practices—likely driven by aesthetic concerns such as fear of tanning or pigmentation, a trend documented in multiple Asian cultural settings[23].

However, these positive attitudes did not always translate into consistent behaviors. The reported practices revealed a preference for physical protection methods and selective sunscreen use, often contingent on season or weather conditions. Protection was most commonly applied during summer or under direct sunlight, despite established evidence that harmful UV exposure can occur throughout the year and even on cloudy days[24]. Very few parents reported reapplying sunscreen regularly or adjusting SPF levels based on environmental conditions, indicating a lack of procedural knowledge and limited integration of sun protection into daily routines. This behavior is not unique to this population; similar inconsistencies have been documented in urban centers across East Asia, where parents express strong beliefs in health protection but struggle to maintain behavioral consistency in everyday practice[25].

Certain demographic patterns also emerged. Parents with higher academic qualifications paradoxically reported less consistent sun protection behavior, a finding that contrasts with studies from Western countries where higher education is often linked to better health practices. This observation is consistent with the multivariate logistic regression results, which showed that higher educational attainment was associated with lower odds of appropriate sun protection practices (OR < 1). This association was identified in the fully adjusted multivariate model, which controlled for relevant covariates that may influence parental sun protection behaviors. One plausible explanation is that these parents may perceive themselves as more informed and thus underestimate the necessity of routine behavioral adherence. Alternatively, work-related time constraints may reduce their capacity for sustained preventive actions[26]. Furthermore, even among those who took some measures, protection was often limited to visible body parts such as the face and neck, with areas like the torso or lower limbs frequently neglected. This partial protection strategy, likely rooted in appearance-related motivations, again points to a need for health messages that emphasize long-term consequences rather than short-term aesthetics.

To address these issues, educational strategies must be reframed to prioritize practical behavior over general awareness. Public health interventions should focus on equipping parents with actionable routines—such as how to apply and reapply sunscreen, when to use protective clothing, and what to consider on overcast days. These programs should include hands-on demonstrations in pediatric clinics or community centers, coupled with clear, culturally adapted guidance materials. Moreover, workplace-based educational outreach targeting both mothers and fathers may help balance gendered caregiving expectations and improve participation. Schools and kindergartens also represent vital access points for modeling sun-safe behavior; structured policies requiring protective measures during outdoor activities could normalize these practices and reduce reliance on parental discretion alone[27,28].

System-level changes are equally important. To sustain behavior over time, parents need environmental support and cues—such as visible reminders, available shade in public areas, and affordable, child-friendly sun protection products. Healthcare professionals must also play a more proactive role. Pediatricians, general practitioners, and nurses should be trained to discuss sun safety as part of routine care, not merely during summer months or dermatologic consultations. Embedding these recommendations into digital health platforms—via appointment reminders, childcare apps, or WeChat-based health messages—may offer low-cost, high-reach strategies for reinforcing behaviors[29,30].

This study has several limitations. First, as a single-center study conducted at a tertiary hospital in Beijing, the findings may not be generalizable to other regions or healthcare settings. Second, the reliance on self-reported data may introduce recall or social desirability bias, potentially affecting the accuracy of responses regarding sun protection behaviors. Third, the cross-sectional design limits the ability to infer causal relationships between knowledge, attitudes, and practices. Additionally, because participants were recruited from dermatology and cosmetic dermatology clinics, the sample may overrepresent parents who are already more attentive to skin health, potentially introducing selection bias and further limiting the generalizability of the findings.

Practical implications

Given the observed associations between knowledge, attitudes, and behaviors, particularly the mediating role of attitude, targeted educational interventions are recommended to enhance parental awareness and promote more consistent protective behaviors in clinical and public health settings. Partnerships with manufacturers to increase access to child-friendly sun-protective clothing and sunscreen formulations may also help address practical barriers such as skin sensitivity and product affordability.

In conclusion, parents in Beijing demonstrated a limited level of knowledge, generally positive attitudes, and suboptimal sun protection practices for their children. The significant associations between knowledge, attitude, and practice suggested that targeted educational interventions aimed at increasing parental awareness and shifting attitudes could enhance sun protection behaviors in both clinical and community settings.

Data availability

All data generated or analysed during this study are included in this published article.

Received: 19 August 2025; Accepted: 11 February 2026
Published online: 18 February 2026

References

1. Health issues of. Ultraviolet tanning appliances used for cosmetic purposes. *Health Phys.* **84**, 119–127. https://doi.org/10.1097/00004032-200301000-00015 (2003).
2. Diehl, K., Thoonen, K., Breitbart, E. W., Pfahlberg, A. B. & Görig, T. Sun Protection and Tanning Behaviors in Caregivers: Prevalence, Determinants, and Associations with Children's Behaviors. *Int. J. Environ. Res. Public. Health*. **19** https://doi.org/10.3390/ijerph19116876 (2022).
3. Li, X. Solar UV radiation and its correlation with skin cancer in Nanning during 1961–2007. *Urban Environ. Urban Ecol.* **22**, 1–6 (2009).
4. Wan, M. et al. Attitudes, Beliefs, and measures taken by parents to protect their children from the sun in Guangzhou City, China. *Photochem. Photobiol.* **92**, 753–759. https://doi.org/10.1111/php.12623 (2016).
5. Shiosaki, J. & Yamamoto, L. G. What are parents reading about the sunscreen controversy? *Pediatr. Dermatol.* **39**, 746–747. https://doi.org/10.1111/pde.15016 (2022).
6. Alamri, G. E. et al. Knowledge and Attitude Regarding the Use of Sun Protection to Prevent Adverse Laser Events Among the General Population in Saudi Arabia: A Cross-Sectional Study. *Cureus.*15, e50157; (2023). https://doi.org/10.7759/cureus.50157
7. Besch-Stokes, J. et al. Skin cancer Knowledge, attitudes and sun protection practices in the Hispanic population: A Cross-Sectional survey. *J. Racial Ethn. Health Disparities.* **10**, 1293–1303. https://doi.org/10.1007/s40615-022-01314-6((2023).
8. Kanellis, V. G., Kanellis, A. L., Von Schuckmann, L. & Miller, A. Education of children, their parents and staff on sun safety practices in the Australian early childhood education setting. *Australas J. Dermatol.* **61**, 160–161. https://doi.org/10.1111/ajd.13219 (2020).
9. Klostermann, S. & Bolte, G. Determinants of inadequate parental sun protection behaviour in their children–results of a cross-sectional study in Germany. *Int. J. Hyg. Environ. Health*. **217**, 363–369. https://doi.org/10.1016/j.ijheh.2013.07.013 (2014).
10. Liu, W., Lai, W. & Wang, X. Investigation of the skin phototypes of Chinese female population. *J. Clin. DERMATOLOGY-NANJING-*. **34**, 420 (2005).
11. Bodekaer Larsen, M. et al. Sun exposure and protection behavior of Danish farm children: parental influence on their children. *Photochem. Photobiol.* **90**, 1193–1198. https://doi.org/10.1111/php.12280 (2014).
12. Littlewood, Z. & Greenfield, S. Parents' knowledge, attitudes and beliefs regarding sun protection in children: a qualitative study. *BMC Public. Health.* **18** https://doi.org/10.1186/s12889-018-5091-8((2018).
13. Thoonen, K., Lima Passos, V., Schneider, F., De Vries, H. & Van Osch, L. Children's sunburn exposed: identification of sun exposure and parental sun protection patterns. *Eur. J. Dermatol.* **31**, 538–548. https://doi.org/10.1684/ejd.2021.4100 (2021).
14. The Skin Beauty Development Working Committee of the Dermatology Branch of the Chinese Medical Association Consensus of Skin Sunscreen Experts. *Chinese Journal of Dermatology.*Volume 50, Pages 316–320; (2017). https://doi.org/10.3760/cma.j.issn.0412-4030.2017.05.002(2017).
15. Lee, F. & Suryohusodo, A. A. Knowledge, attitude, and practice assessment toward COVID-19 among communities in East Nusa Tenggara, indonesia: A cross-sectional study. *Front. Public. Health.* **10**, 957630. https://doi.org/10.3389/fpubh.2022.957630 (2022).
16. Liao, J., Scholes, S., Mawditt, C., Mejía, S. T. & Lu, W. Comparing relationships between health-related behaviour clustering and episodic memory trajectories in the United States of America and England: a longitudinal study. *BMC public health.*22, 1367; (2022).
17. Dighe, S. et al. Healthier school food and physical activity environments are associated with lower student body mass index. *Prev. Med. Rep.* **19**, 101115 (2020).
18. Nagelhout, E. et al. (ed, S.) Differences in reported sun protection practices, skin cancer knowledge, and perceived risk for skin cancer between rural and urban high school students. *Cancer Causes Control* **30** 1251–1258 https://doi.org/10.1007/s10552-019-01228-5 (2019).
19. Joschko, L., Pálsdóttir, A. M., Grahn, P. & Hinse, M. Nature-Based therapy in individuals with mental health Disorders, with a focus on mental Well-Being and connectedness to Nature-A pilot study. *Int. J. Environ. Res. Public. Health*. **20** https://doi.org/10.3390/ijerph20032167 (2023).
20. Nagelhout, E. S. et al. Differences in reported sun protection practices, skin cancer knowledge, and perceived risk for skin cancer between rural and urban high school students. *Cancer Causes Control*. **30**, 1251–1258 (2019).
21. Littlewood, Z. & Greenfield, S. Parents' knowledge, attitudes and beliefs regarding sun protection in children: a qualitative study. *BMC public. Health*. **18**, 1–11 (2018).
22. Li, Q. *Preparedness and Opinions of Nurses Towards Intimate Partner Violence in China: A Mixed Methods Study* (Johns Hopkins University, 2023).

23. Loh, C. H., Tang, P. Y. & Lee, H. Y. Erythematous papular eruption of acute onset in a 51-year-old male. *International J. Dermatology* **60**; (2021).
24. Mirza, F. N. et al. Treatment, survival and prognostic factors of cutaneous mucoepidermoid carcinoma: A distinct entity with indolent clinical course. *J. Am. Acad. Dermatol.* **83**, 1827 (2020).
25. Joschko, L., Palsdottir, A. M., Grahn, P. & Hinse, M. Nature-based therapy in individuals with mental health disorders, with a focus on mental well-being and connectedness to nature—a pilot study. *International journal of environmental research and public health*.20, 2167; (2023).
26. Gao, Y. et al. The impact of parental migration on health status and health behaviours among left behind adolescent school children in China. *BMC public. Health*. **10**, 1–10 (2010).
27. Jylänki, P., Mbay, T., Hakkarainen, A., Sääkslahti, A. & Aunio, P. The effects of motor skill and physical activity interventions on preschoolers' cognitive and academic skills: A systematic review. *Prev. Med.* **155**, 106948 (2022).
28. Obita, G. & Alkhatib, A. Effectiveness of lifestyle nutrition and physical activity interventions for childhood obesity and associated comorbidities among children from minority ethnic groups: A systematic review and meta-analysis. *Nutrients*.15, 2524; (2023).
29. Tang, H. et al. The prognostic role of WeChat-based instant doctor–patient communication in patients with advanced pancreatic cancer. *J. Cancer Surviv.* **18**, 1903–1911 (2024).
30. Wan, M. et al. Attitudes, beliefs, and measures taken by parents to protect their children from the sun in Guangzhou City, China. *Photochem. Photobiol.* **92**, 753–759 (2016).

Acknowledgements
Not applicable.

Author contributions
Xuefei Lin and Ting Li carried out the studies, participated in collecting data, and drafted the manuscript. Xiao Chang performed the statistical analysis, or interpretation of data and participated in its design. All authors read and approved the final manuscript.

Funding
None.

Declarations

Ethics approval and consent to participate
All procedures were performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments. The study population comprised parents of children, and all procedures were approved by the Ethics Committee of Xuanwu Hospital, Capital Medical University (MR-11-25-047140). Written informed consent was obtained from all participants prior to data collections. The study was carried out in accordance with the applicable guidelines and regulations.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-40202-5.

Correspondence and requests for materials should be addressed to X.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026